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CHENNAI

22BAE1268

Final Assessment Test(FAT) - Apr/May 2025

Programme	**B.Tech.**	Semester	**Winter Semester 2024-25**
Course Code	**BCSE309L**	Faculty Name	**Prof. Jannath Nisha O S**
Course Title	**Cryptography and Network Security**	Slot	**F1+TF1**
		Class Nbr	**CH2024250502346**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: To know the fundamental mathematical concepts related to security.
CO2: To understand concept of various cryptographic techniques.
CO3: To apprehend the authentication and integrity process of data for various applications
CO4: To know fundamentals of Transport layer security, web security, E-Mail Security and IP Security

Answer all Questions (10 × 10 Marks)

01. A military parade is arranged for a group of soldiers. The army chief commander observes the following: when the soldiers are lined up in rows of 5, 3 soldiers are left over; when lined up in rows of 7, 1 soldier is left over; and when lined up in rows of 8, 6 soldiers are left over. Using the Chinese Remainder Theorem, determine the minimum number of soldiers needed (integer X) that satisfies these conditions.

[10] (CO1/K2)

02. Consider a simplified RC4 stream cipher with a state vector S of size 4 x 2 bits. Since S can take values from 0 to 3, you will operate on 2 bit plaintext at a time. Encrypt the given plaintext P = [0,3,1,1] step by step using the key K = [2,0].

[10] (CO2/K3)

03. A company, SecureCorp has implemented RSA encryption to safeguard its internal communications. The company's IT team has selected a public key with an exponent e = 11 and a modulus n = 119.
a. Riya, an employee, intends to send a confidential message to her colleague Arjun. After converting her message into numeric form, she obtains the value M = 45. determine the encrypted message that Riya will send to Arjun. (3 marks)

b. Later, Arjun receives the encrypted message. Using his private key, which is known only to him, he computes the original message sent by Riya. (7 marks)

[10] (CO2/K3)

04. A company wants to ensure the integrity of its messages by using the MD5 hash algorithm. The IT team is tasked with processing messages of different lengths and understanding how MD5 works in detail.
a. Compute the value of the padding field and length field in MD5 if the length of the message is (4 Marks)
   i. 1470 bits
   ii. 449 bits

b. Explain the working of one round operation of MD5 hash algorithm. (6 Marks)

[10] (CO3/K1)

05. Emma and Jack are two security researchers working on a lightweight cryptographic protocol for IoT devices. To ensure secure communication, they decide to use Elliptic Curve Cryptography. To test the set up they choose a simple elliptic curve defined over a small prime field: $E_{17}(1, 6)$: $y^2 = x^3+x+6 \bmod 17$ with a generator G = (1, 5). Jack chooses his private key $K_J = 4$.
a. Compute Jack's public key. (4 Marks)

b. Emma chooses random value $K_E = 2$ to encrypt the message M = (2, 13). Determine the ciphertext. (6 Marks)

[10] (CO2/K4)

06. a. As a cybersecurity analyst working for a company that uses the ElGamal digital signature scheme, you are responsible for verifying its authenticity. You receive a signed document with the following details. Signature: (5,9), Hash value of message:7 and Public Key components of the sender(prime number, primitive root, public key):(17,3,10).Determine whether the signature generated is authentic. Show the complete verification process. (3Marks).

b. Assume you are using ElGamal digital signature with parameters (prime number, primitive root, private key): (13,2,5)

i)Calculate the public key.(2 Marks)

ii)Compute the El gamal signature for message with hash value 9 and a random integer 7.(3 Marks)

iii)Verify if the generated signature is valid.(2 Marks)

[10] (CO3/K4)

07. The following are core security requirements in Kerberos Version 4.

a. The Ticket Granting Server (TGS) or an application service must be able to verify that the user presenting a ticket is the same user to whom the ticket was issued.(5 Marks)

b. Servers must authenticate themselves to users. (5 Marks)

Explain how the above requirements are addressed in Kerberos Version 4, and justify your answer with the relevant authentication dialogues.

[10] (CO3/K4)

08. Jack, a cybersecurity engineer, works remotely and often logs into his company's private cloud servers to monitor security logs and manage firewall rules. To ensure a secure connection, his company uses encrypted protocols for authentication and data exchange. However, Jack is reviewing the security policies and wants to understand whether the network should rely on SSL or TLS for a more secure connection.

a. Explain the handshake protocol in SSL with a neat diagram, including its key steps and how it ensures secure key exchange.(6 Marks)

b. Draw the TLS protocol stack and explain the significance of Heartbeat protocol used in TLS (4 Marks)

[10] (CO4/K5)

09. Alexa is a journalist working for an international news agency. She often communicates with whistle blowers and sources who provide sensitive information. To ensure that her emails and documents remain confidential, authentic, and tamper-proof, she decides to use Pretty Good Privacy (PGP) for secure communication.

a. Explain how Alexa can use PGP to provide authentication , confidentiality and both. (6 Marks)

b. Explain about PGP message generation and reception with a neat diagram. (4 Marks)

[10] (CO4/K4)

10. A financial organization aims to enhance its cybersecurity by deploying an Intrusion Detection System (IDS). They require a solution that can monitor network traffic for threats such as DDoS attacks, detect abnormal activities on individual servers, such as unauthorized file access, and actively block malicious traffic in real time.

a. Classify and explain the different types of intrusion detection system. (7 Marks)

b. Based on your IDS classification, recommend the most suitable types of IDS, the financial organization can implement for each requirement, and justify your choice. (3 Marks)

[10] (CO4/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)